

Fisher Information: From Classical Statistics to Quantum Neural Networks

Mathematical Foundations, the QFI Matrix, and Optimisation Geometry

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May 6, 2026

Outline I

- 1 Classical Fisher Information
- 2 Quantum Fisher Information
- 3 The QFI Matrix in QNNs
- 4 Natural Gradient and TDVP
- 5 Effective Dimension and Expressivity
- 6 Barren Plateaus and the QFI
- 7 Summary

Statistical Model and the Score Function

Parametric probability distribution $p(x|\theta)$, $\theta \in \mathbb{R}^d$.

Score function: $s(x; \theta) = \partial_\theta \log p(x|\theta)$.

Key identity from $\int p dx = 1$: $\mathbb{E}_{x \sim p}[s(x; \theta)] = 0$. The score has zero mean; its variance is the Fisher information.

Goals

- Estimate θ from data
- Quantify sensitivity of p to θ
- Define a geometry on parameter space

Intuition

High sensitivity $\Rightarrow$ nearby distributions are distinguishable $\Rightarrow$ more information about θ in a single observation.

Definition and Equivalent Forms

Fisher Information (scalar θ):

$$\mathcal{I}(\theta) = \mathbb{E}_{x \sim p} \left[\left(\partial_{\theta} \log p(x|\theta) \right)^2 \right] = \text{Var}[s(x; \theta)].$$

Equivalent form:

$$\mathcal{I}(\theta) = -\mathbb{E} \left[\partial_{\theta}^2 \log p(x|\theta) \right].$$

Matrix version for $\theta \in \mathbb{R}^d$:

$$\mathcal{I}_{jk}(\theta) = \mathbb{E} \left[\frac{\partial \log p}{\partial \theta_j} \frac{\partial \log p}{\partial \theta_k} \right] = -\mathbb{E} \left[\frac{\partial^2 \log p}{\partial \theta_j \partial \theta_k} \right].$$

Derivation sketch

Differentiate $\mathbb{E}[\partial_{\theta} \log p] = 0$ once more: $\mathbb{E}[\partial_{\theta}^2 \log p] + \mathbb{E}[(\partial_{\theta} \log p)^2] = 0$, giving the equivalence immediately.

Cramér–Rao bound: for any unbiased $\hat{\theta}$,

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)}.$$

Matrix version: $\text{Cov}(\hat{\theta}) \geq \mathcal{I}(\theta)^{-1}$ (Löwner order).

- $\mathcal{I}$ sets a fundamental precision limit
- High $\mathcal{I} \Rightarrow$ sharper inference
- MLEs saturate the bound asymptotically

Natural gradient (Amari 1998): steepest descent on the manifold $(\mathbb{R}^d, \mathcal{I})$:

$$\theta_{t+1} = \theta_t - \eta \mathcal{I}(\theta)^{-1} \nabla_{\theta} L.$$

Fisher–Rao metric

$$ds^2 = \sum_{jk} \mathcal{I}_{jk} d\theta_j d\theta_k.$$

Reparameterisation-invariant; steepest descent on information-geometric manifold.

From Classical to Quantum

Classical: $p(x|\theta) \rightarrow$ density matrix $\rho(\theta)$.

Pure variational state:

$$|\psi(\theta)\rangle = U(\theta)|0\rangle^{\otimes n}, \quad \rho = |\psi\rangle\langle\psi|.$$

Observable expectation:

$$f(\theta) = \langle\psi(\theta)|\hat{O}|\psi(\theta)\rangle.$$

SLD (Symmetric Logarithmic Derivative):

$$\partial_\theta \rho = \frac{1}{2}(L_\theta \rho + \rho L_\theta).$$

QFI (mixed state): $F_Q = \text{Tr}[\rho L_\theta^2]$.

Quantum Cramér–Rao bound

For any measurement strategy:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{F_Q(\theta)}.$$

$F_Q \geq \mathcal{I}_{\text{classical}}$ for every measurement. The QFI maximises over all quantum measurements.

Key difference

The classical FI depends on a measurement choice. The QFI is the *maximum* over all possible quantum measurements—the ultimate precision limit.

QFI for Pure States and the Generator Formula

For $|\psi(\theta)\rangle$ (single parameter):

$$F_Q(\theta) = 4 \left(\langle \partial_\theta \psi | \partial_\theta \psi \rangle - |\langle \psi | \partial_\theta \psi \rangle|^2 \right).$$

Generator form. If $|\psi(\theta)\rangle = e^{-i\theta G}|\psi_0\rangle$ with Hermitian generator G ($G^\dagger = G$), then $\partial_\theta |\psi\rangle = -iG|\psi\rangle$ and:

$$F_Q = 4 \text{Var}_{|\psi\rangle}(G) = 4(\langle G^2 \rangle - \langle G \rangle^2).$$

Example: single qubit. $|\psi(\theta)\rangle = e^{-i\theta\sigma_y/2}|0\rangle$,
 $G = \sigma_y/2$, eigenvalues $\pm\frac{1}{2}$:

$$F_Q = 4 \text{Var}(\sigma_y/2) = 4 \cdot \frac{1}{4} = 1.$$

Uniform sensitivity; no barren plateau.

Metrological scaling: for N separable qubits
 $F_Q \leq N$ (SQL); with entanglement $F_Q \sim N^2$
(Heisenberg limit).

Interpretation

$F_Q = 4 \text{Var}(G)$: large generator fluctuations mean the state is highly sensitive to θ ; information about θ is efficiently encoded. Maximising $\text{Var}(G)$ is the task of quantum metrology.

The QFI Matrix: Definition and Geometric Meaning

For a variational state $|\psi(\boldsymbol{\theta})\rangle$, $\boldsymbol{\theta} \in \mathbb{R}^p$, the QFI is a *real symmetric* $p \times p$ matrix:

$$F_{jk}(\boldsymbol{\theta}) = 4 \operatorname{Re} \left(\langle \partial_j \psi | \partial_k \psi \rangle - \langle \partial_j \psi | \psi \rangle \langle \psi | \partial_k \psi \rangle \right)$$

$$|\partial_j \psi\rangle \equiv \partial |\psi(\boldsymbol{\theta})\rangle / \partial \theta_j.$$

Geometric meaning. $\mathbf{F}$ is the **Fubini–Study metric** on the variational manifold $\mathcal{M}$:

$$ds^2 = \sum_{jk} F_{jk} d\theta_j d\theta_k.$$

Measures information-geometric distance between neighbouring quantum states in $\mathcal{H}$.

Covariance form (product-of-rotations circuit $U(\boldsymbol{\theta}) = \prod_j e^{-i\theta_j P_j/2}$):

$$F_{jk} = 4 \operatorname{Cov}_{|\psi\rangle}(G_j, G_k), \quad G_j = P_j/2.$$

Circuit formula (two-fold shifts)

State-overlap function: $\mathcal{O}(\boldsymbol{\theta}) = |\langle \psi_{\text{ref}} | \psi(\boldsymbol{\theta}) \rangle|^2$.

$$F_{jk} = \frac{1}{2} [\mathcal{O}_{j+,k+} - \mathcal{O}_{j+,k-} - \mathcal{O}_{j-,k+} + \mathcal{O}_{j-,k-}],$$

$\mathcal{O}_{j\pm,k\pm}$: shift $\theta_j \rightarrow \theta_j \pm \frac{\pi}{2}$, $\theta_k \rightarrow \theta_k \pm \frac{\pi}{2}$. Cost: $\mathcal{O}(p^2)$ circuit evaluations.

Derivation of the Covariance Form

For $U(\boldsymbol{\theta}) = \prod_j e^{-i\theta_j P_j/2}$ in the single-layer case: $\partial_j |\psi\rangle = -\frac{i}{2} P_j |\psi\rangle$.

Compute each term in the QFI formula:

$$\begin{aligned}\langle \partial_j \psi | \partial_k \psi \rangle &= \frac{1}{4} \langle \psi | P_j P_k | \psi \rangle, \\ \langle \partial_j \psi | \psi \rangle &= \frac{i}{2} \langle P_j \rangle, \quad \langle \psi | \partial_k \psi \rangle = -\frac{i}{2} \langle P_k \rangle.\end{aligned}$$

Insert and take the real part:

$$F_{jk} = 4 \operatorname{Re} \left(\frac{1}{4} \langle P_j P_k \rangle - \frac{i}{2} \langle P_j \rangle \cdot \frac{i}{2} \langle P_k \rangle \right) = \langle P_j P_k \rangle - \langle P_j \rangle \langle P_k \rangle = 4 \operatorname{Cov}(G_j, G_k). \quad \square$$

Multi-layer extension

For L -layer circuits, commutators between generators of different layers contribute. The two-fold shift circuit formula handles all layers exactly at cost $O(p^2)$.

Parameter-Shift Rule and the QFI Connection

Parameter-shift rule for $f(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | \hat{O} | \psi(\boldsymbol{\theta}) \rangle$ with generator eigenvalues $\pm \frac{1}{2}$:

$$\frac{\partial f}{\partial \theta_j} = \frac{1}{2} \left[f\left(\theta_j + \frac{\pi}{2}\right) - f\left(\theta_j - \frac{\pi}{2}\right) \right].$$

Exact, no approximation; 2 circuit evaluations per parameter.

Commutator form: $\partial_{\theta_j} f = \langle \psi | i[G_j, \hat{O}] | \psi \rangle$.

Diagonal QFI entry: $F_{jj} = 4 \text{Var}_{|\psi\rangle}(G_j)$.

Cauchy–Schwarz bound (gradient–QFI)

$$\left| \frac{\partial f}{\partial \theta_j} \right|^2 \leq F_{jj} \cdot \text{Var}_{|\psi\rangle}(\hat{O}).$$

Small generator variance $\Rightarrow$ small $F_{jj} \Rightarrow$ all gradients of all observables are small $\Rightarrow$ barren plateau for parameter θ_j .

The **Quantum Geometric Tensor (QGT)**:

$$\chi_{jk} = \langle \partial_j \psi | \partial_k \psi \rangle - \langle \partial_j \psi | \psi \rangle \langle \psi | \partial_k \psi \rangle.$$

Decomposition: $\chi_{jk} = \frac{1}{4}F_{jk} + i\Omega_{jk}$,

- $F_{jk} = 4 \operatorname{Re}(\chi_{jk})$: **QFI matrix** (Fubini–Study metric)
- $\Omega_{jk} = \operatorname{Im}(\chi_{jk})$: **Berry curvature** (geometric phase)

Geometric significance. χ is the pullback of the Fubini–Study form on $\mathbb{P}(\mathcal{H})$ via $\theta \mapsto |\psi(\theta)\rangle$. It unifies information geometry (real part) and symplectic geometry (imaginary part).

Connection to Berry phase

The Berry phase around a loop γ in parameter space is:

$$\varphi_B = \oint_{\gamma} \operatorname{Im} \langle \psi | d | \psi \rangle = \iint_{\Sigma} \Omega_{jk} d\theta_j \wedge d\theta_k.$$

Natural Gradient in Quantum Circuits

Natural gradient (Amari 1998; Stokes et al. 2020):

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \mathbf{F}(\boldsymbol{\theta})^+ \nabla_{\boldsymbol{\theta}} C.$$

$\mathbf{F}^+$: Moore–Penrose pseudoinverse (needed when $\mathbf{F}$ is rank-deficient).

Derivation. Minimise $C(\boldsymbol{\theta} + \delta\boldsymbol{\theta})$ subject to fixed Bures distance:

$$\delta\boldsymbol{\theta}^\top \mathbf{F} \delta\boldsymbol{\theta} = \varepsilon^2.$$

Lagrange multipliers give $\delta\boldsymbol{\theta} \propto -\mathbf{F}^{-1} \nabla C$.

The update is *reparameterisation-invariant*: it always descends the steepest direction on $\mathcal{M}$.

TDVP equations of motion

Dirac–Frenkel variational principle: project $(i\partial_t - H)|\psi\rangle = 0$ onto $\mathcal{T}_{\boldsymbol{\theta}}\mathcal{M} = \text{span}\{|\partial_j\psi\rangle\}$:

$$\sum_k F_{jk} \dot{\theta}_k = 2 \text{Im} \langle \partial_j \psi | H | \psi \rangle.$$

This is the Time-Dependent Variational Principle for VQAs.

Imaginary-time TDVP = natural gradient VQE

Substituting $t \rightarrow -i\tau$: $\dot{\boldsymbol{\theta}} = -\mathbf{F}^{-1} \nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta})$.
Equivalent to **Stochastic Reconfiguration** in VMC.

Computing the Natural Gradient and SR Connection

Exact cost: $O(p^2)$ circuit evaluations plus $O(p^3)$ matrix inversion. Expensive for $p \sim 10^3$.

Practical approximations:

- 1 **Diagonal:** use only F_{jj} ; $O(p)$ evaluations.
- 2 **Block-diagonal:** group by layer; $O(p^2/L)$ cost.
- 3 **Regularised:** replace $\mathbf{F}$ by $\mathbf{F} + \lambda I$.

SR connection (Sorella 1998):

$$\theta_j \leftarrow \theta_j - \eta \sum_k S_{jk}^{-1} g_k,$$

with $S_{jk} = \langle \partial_j \psi | \partial_k \psi \rangle - \langle \partial_j \psi | \psi \rangle \langle \psi | \partial_k \psi \rangle = F_{jk}/4$.
SR is the natural gradient w.r.t. the Fubini–Study metric.

Regularised update rule

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta (\mathbf{F} + \lambda I)^{-1} \nabla_{\boldsymbol{\theta}} C.$$

- $\lambda \rightarrow 0$: pure natural gradient
- $\lambda \rightarrow \infty$: standard gradient
- $\lambda = \|\nabla C\|$: adaptive

Quantum advantage

The QFI matrix elements F_{jk} can be measured on a quantum device via the two-fold shift formula, without classical density-matrix simulation.

Effective Dimension via the QFI

Abbas et al. (2021): define the **effective dimension** via the average log-volume of the QFI:

$$d_{\text{eff}} = \frac{2}{\log N} \log \left[\frac{1}{V} \int_{\boldsymbol{\theta}} \sqrt{\det \left(I + \frac{N}{2\pi \ln N} \mathbf{F}(\boldsymbol{\theta}) \right)} d\boldsymbol{\theta} \right],$$

where N is the number of training samples.

Properties

- High d_{eff} : large effective capacity.
- Some QNNs exceed d_{eff} of classical networks at equal parameter count.
- $\mathbf{F} \approx 0$: redundant parameters; flat loss directions.
- $d_{\text{eff}} \rightarrow 0$ as $\mathbf{F} \rightarrow 0$: model collapses.
- Maximal expressibility (Haar) gives high d_{eff} *but also barren plateaus*.

QFI as trainability indicator

Cauchy–Schwarz on the Fubini–Study metric:

$$\left| \frac{\partial C}{\partial \theta_j} \right|^2 \leq \lambda_{\max}(\mathbf{F}) \cdot \text{Var}_{|\psi\rangle}(\hat{B}).$$

Small $\lambda_{\max}(\mathbf{F}) \Leftrightarrow$ all gradients small $\Leftrightarrow$ barren plateau. Monitor $\mathbf{F}$ spectrum as an early-warning diagnostic.

QFI Rank, Redundant Parameters, and Expressibility

Rank deficiency. If $\text{rank}(\mathbf{F}) < p$, there exist directions $\mathbf{v}$ with $\mathbf{v}^\top \mathbf{F} \mathbf{v} = 0$: moving along $\mathbf{v}$ does not change the quantum state $\Rightarrow$ *redundant parameters*.

Consequences:

- Flat loss directions (pseudo-barren)
- Can aid optimisation (escape local minima)
- Inflate circuit depth needlessly
- Use $\text{rank}(\mathbf{F})$ to prune circuits

Expressibility (Meyer et al. 2021): measures how uniformly the circuit covers $U(2^n)$. High expressibility $\Leftrightarrow$ large $\det(\mathbf{F})$ on average $\Leftrightarrow$ high d_{eff} .

The expressibility–trainability trade-off

Circuit	d_{eff}	Trainable?
Shallow, structured	Low	Yes
Medium depth	Medium	Often
Deep random	High	No (BP)

Optimal VQAs balance expressibility with structured ansätze that avoid the Haar-random concentration of measure.

Barren Plateaus: Definition and Mathematical Origin

Definition (McClean et al. 2018)

A circuit family $\{U(\boldsymbol{\theta})\}$ has a **barren plateau** if:

$$\mathbb{E}_{\boldsymbol{\theta}}[\partial_{\theta_j} C] = 0, \quad \text{Var}_{\boldsymbol{\theta}}[\partial_{\theta_j} C] = O(b^{-n}), \quad b > 1.$$

Mathematical origin. For Haar-random U on $\text{SU}(2^n)$:

$$\mathbb{E}_U[\langle \hat{O} \rangle] = \frac{\text{Tr} \hat{O}}{2^n}, \quad \text{Var}[\partial_{\theta_j} C] \sim \frac{\text{Tr} \hat{O}^2}{4^n}.$$

For $\hat{O} = Z_k$: $\text{Var} \sim 1/2^n$. Concentration of measure (Lévy's lemma): functions concentrate near their mean with prob. $\geq 1 - 2e^{-\Omega(2^{2n} \epsilon^2)}$.

Consequence

Deep random circuits: $C(\boldsymbol{\theta}) \approx \text{Tr} \hat{O}/2^n$ almost everywhere. Flat landscape. *Not* a local-minima problem.

Direct QFI link

Cauchy-Schwarz: $\text{Var}[\partial_{\theta_j} C] \leq \frac{1}{4} F_{jj} \text{Var}(\hat{O})$.

For Haar-random circuits: $F_{jj} \sim 1/2^n \rightarrow 0$.

Therefore: $F_{jj} \rightarrow 0 \iff$ barren plateau.

Monitoring F_{jj} before and during training gives an *early warning* of trainability collapse.

Noise-Induced Barren Plateaus and Mitigation

Hardware noise (Wang et al. 2021): depolarising noise rate ε per gate:

$$|\langle \hat{O} \rangle_{\text{noisy}}| \leq (1 - \varepsilon)^{pd} |\langle \hat{O} \rangle_{\text{ideal}}|,$$

$$\text{Var}[\partial_{\theta_j} C_{\text{noisy}}] \sim (1 - \varepsilon)^{2pd} \text{Var}[\partial_{\theta_j} C_{\text{ideal}}].$$

Affects even *shallow* circuits; cannot be fixed by initialisation.

Mitigation strategies

Strategy	Mechanism
Local cost function	$\text{poly}(n)$ gradient scaling
Problem ansatz	exploit symmetries
Layer-wise training	$O(1)$ depth locally
Identity initialisation	$O(1)$ initial gradients
Natural gradient	QFI-respecting steps
Shallow circuits	avoid Haar regime

Physics-inspired ansätze

Circuits mirroring Hamiltonian structure (QPINNs) naturally avoid the Haar regime and mitigate BPs without additional overhead.

Entanglement entropy and trainability.

$$S(\rho_A) = -\text{Tr}(\rho_A \log \rho_A):$$

Regime	$S(\rho_A)$	Training
Low	≈ 0	Classically simulable
Volume-law	$\Theta(n)$	Barren plateaus
Area-law	$O(\partial A)$	Trainable

Area-law (shallow circuits, MERA) is the sweet spot for VQA trainability.

Quantum Neural Tangent Kernel (QNTK)

$$\Theta(\mathbf{x}, \mathbf{x}') = \sum_j \frac{\partial f(\mathbf{x}; \boldsymbol{\theta})}{\partial \theta_j} \frac{\partial f(\mathbf{x}'; \boldsymbol{\theta})}{\partial \theta_j}.$$

- Governs linearised training dynamics
- At a barren plateau: $\Theta \approx 0$ for all $\mathbf{x}, \mathbf{x}'$
- Bound: $\Theta(\mathbf{x}, \mathbf{x}) \leq \lambda_{\max}(\mathbf{F})$
- Connects QNN training to quantum kernel SVMs

Key Equations at a Glance

Concept	Formula
Classical FI	$\mathcal{I}(\theta) = \mathbb{E}[(\partial_\theta \log p)^2] = -\mathbb{E}[\partial_\theta^2 \log p]$
Cramér–Rao	$\text{Var}(\hat{\theta}) \geq 1/\mathcal{I}(\theta)$
Classical nat. grad.	$\theta \leftarrow \theta - \eta \mathcal{I}^{-1} \nabla L$
QFI matrix	$F_{jk} = 4 \text{Re}(\langle \partial_j \psi \partial_k \psi \rangle - \langle \partial_j \psi \psi \rangle \langle \psi \partial_k \psi \rangle)$
Covariance form	$F_{jk} = 4 \text{Cov}_{ \psi\rangle}(G_j, G_k)$
Generator variance	$F_Q = 4 \text{Var}_{ \psi\rangle}(G)$
Fubini–Study metric	$ds^2 = \sum_{jk} F_{jk} d\theta_j d\theta_k$
Quantum geometric tensor	$\chi_{jk} = \frac{1}{4} F_{jk} + i\Omega_{jk}$
Parameter-shift	$\partial_{\theta_j} f = \frac{1}{2}[f(\theta_j + \frac{\pi}{2}) - f(\theta_j - \frac{\pi}{2})]$
Gradient–QFI bound	$ \partial_{\theta_j} C ^2 \leq \lambda_{\max}(\mathbf{F}) \cdot \text{Var}(\hat{O})$
Natural grad. (quantum)	$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \mathbf{F}^+ \nabla C$
TDVP	$\sum_k F_{jk} \dot{\theta}_k = 2 \text{Im} \langle \partial_j \psi H \psi \rangle$
Barren plateau	$\text{Var}[\partial_{\theta_j} C] \sim 1/2^n, F_{jj} \sim 1/2^n$

Classical Fisher information

- Variance of the score function
- Cramér–Rao precision limit
- Fisher–Rao Riemannian metric
- Natural gradient: reparameterisation-invariant descent

Quantum generalisation

- QFI = maximum FI over all measurements
- Pure-state formula via state overlaps
- QFI matrix = Fubini–Study metric on $\mathcal{M}$
- Covariance of generators; exact circuit formula
- QGT: QFI (metric) + Berry curvature (symplectic)

QFI in QNN optimisation

- Natural gradient = $\mathbf{F}^{-1}\nabla C$: geometry-aware, invariant
- TDVP: imaginary-time = natural gradient
VQE = stochastic reconfiguration
- Effective dimension via $\det(\mathbf{F})$: quantifies model capacity
- Barren plateaus: $F_{jj} \sim 1/2^n$; QFI spectrum is an early-warning diagnostic

Central message

The QFI is simultaneously the *optimisation metric*, the *expressivity measure*, and the *barren-plateau detector* for quantum neural networks.

- ➊ **Score identity.** Verify $\mathbb{E}[\partial_\theta \log p] = 0$ for $p(x|\theta) = \mathcal{N}(x; \theta, 1)$; compute $\mathcal{I}(\theta)$.
- ➋ **QFI of a single qubit.** For $|\psi(\theta)\rangle = R_y(\theta)|0\rangle$, compute F_{11} via the Fubini–Study formula and via $4 \text{Var}(G)$. Verify $F_{11} = 1$.
- ➌ **Two-parameter covariance form.** For $U(\theta_1, \theta_2) = e^{-i\theta_1 Z/2} e^{-i\theta_2 X/2}$, compute the full 2×2 matrix $F_{jk} = 4 \text{Cov}(G_j, G_k)$.
- ➍ **Circuit formula verification.** Verify the two-fold shift formula for F_{12} numerically for the above circuit on $|0\rangle$.
- ➎ **Barren plateau scaling.** Estimate $\text{Var}[\partial C / \partial \theta_j]$ for $n = 2, \dots, 8$ qubit random circuits of depth $2n$. Compare with diagonal F_{jj} .
- ➏ **Natural gradient vs. Adam.** Implement both for a two-qubit VQE problem; compare convergence curves.

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